

A broadband and low-frequency sound absorber of sonic black holes with multi-layered micro-perforated panels

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ARTICLE INFO

Keywords:

Sonic black hole
Micro-perforated panel
Transfer matrix method
Sound absorption
Broadband and low-frequency

ABSTRACT

Sonic black hole (SBH) absorber is a promising strategy in noise reduction. However, low-frequency sound waves are difficult to be captured and dissipate due to the terminal truncation and the discontinuity of the admittance for the actual SBH. To make up the disadvantages, this paper incorporates micro-perforated panels (MPP) into the SBH to improve its low-frequency property. The design is realized by inserting the multi-layered MPPs (MLP) into the rings with decaying inner radius inside a SBH duct, termed the MLP-SBH duct. The theoretical and numerical results show that the proposed absorber can achieve effective sound absorption in the frequencies above 225 Hz, in which the thickness of the absorber is only $\lambda/15$ with the λ being the corresponding wavelength. Mechanism studies indicate that the specific slow wave effect of the SBH can effectively mitigate the requirements for the absorber length. Closer examination reveals that the high acoustic resistance characteristics of the MLP remarkably compensate for the acoustic resistance mismatch of the SBH in low-frequency range. Thus, attributing to the interplay of the SBH and MLP effects, the proposed absorber achieves broadband and low-frequency sound absorption. Based on the deduction and validation of the transfer matrix method (TMM), the influence of the structural parameters on the sound absorption is investigated. The analysis indicates that the absorption performance of the proposed absorber barely depends on the damping and the number of rings, which allows the MLP-SBH duct to maintain the excellent performance without abundant rings and extra absorbent material. The proposed strategy is an inspiration for the application of the SBH-based structures.

1. Introduction

Effective broadband and low-frequency sound absorption has always been a challenge in noise control engineering. Conventional resonant absorption materials need a deep air cavity to achieve ideal sound absorption, resulting in a bulky structure [1–3]. To overcome these deficiencies, researchers have been committed to exploring new sound absorption materials/structures. Acoustic metamaterials [4–7] consisting of periodic structures might be a novel solution, which are expected to achieve perfect sound absorption with sub-wavelength size. Another promising approach is to increase the effective length of the structure through a coiled or labyrinthine absorber. Researches [8–10] have shown that sound waves will slow down in these ingenious proposals, thereby reducing the size of the structure in another way and absorbing the low-frequency sound, formally named slow wave effect. However, it is difficult for these proposals to achieve both broadband and low-

frequency sound absorption effectively.

The Acoustic Black Hole (ABH) is an absorber for passive vibration and noise control that was recently developed [11–14]. The technique was firstly presented in Vibration community by Mironov [15,16]. In his work the flexural waves will be trapped in a tapered wedge or a circular indentation having a power-law decreasing thickness profile. To be precise, these tapered structure that absorb bending waves should be called “Vibrational Black Hole (VBH)”. However, in practice, it is impossible to reduce the thickness of the structure terminate into zero [17]. In other words, there is inevitably a truncation in the terminate of the VBH. The truncation incurs drastic damage on the absorption performance of the VBH to waves, especially low-frequency waves. This result in a cut-on frequency (f_{cut-on}) for the VBH, above which the counterpart exhibits the “black hole” property, and below which the VBH effect will be greatly diminished [18]. Numerous strategies have been explored to drop the cut-on frequency f_{cut-on} . For the VBH beam, it

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<https://doi.org/10.1016/j.apacoust.2023.109817>

Received 29 August 2023; Received in revised form 9 December 2023; Accepted 12 December 2023

Available online 11 January 2024

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was found that attaching a damping layer at the terminate of the tapered wedge can mitigate the adverse influence given by the truncation [19]. In addition, many configurations of the VBH have been proposed to improve the effect in low-frequency range, including spiral VBH beams [20–22], symmetric VBH beams [23,24], segmented VBH beams [25,26] and elliptic plates with VBH [27]. Besides, metamaterials are also used to elevate low-frequency wave absorption because of the characteristic of bandgaps. It is convenient for periodic VBHs to control the attenuation band by capitalizing on the Bragg scattering effect [28–31]. Another prospective strategy is based on attaching a negative stiffness element to the VBH beam, which hopefully provides broadband vibration attenuation [32]. More brief information about the VBH can be achieved in the recent review paper [33].

For sound waves, the concept of the retarding duct was proposed by Mironov [34], inside which a set of rings with a power-law varying inner radius are installed in a one-dimensional waveguide. Similarly, the group and phase velocity of incident sound wave in the retarding duct will be reduced into zeros at the terminate. So the sound waves can be trapped in it, hence achieving a zero reflection coefficient of sound pressure. The term of this retarding duct is “Sonic Black Hole (SBH)” to distinguish from VBH, both are classified as ABH. Comparatively, SBH is much less exploited. It was 15 years after the SBH duct was first presented that scholars resumed study on it. Guasch et al. [35] derived the analytical solution of governing equation and transfer matrix method (TMM) for the SBH duct, which provided an efficient solution for the reflection coefficient of sound pressure. This work also explored the influence of structure parameters on the reflection coefficient. In his recent research, the SBH effect has been investigated by analogy with the governing equations of a *meta-fluid* [36]. In addition, fractional order operators have also been applied to estimate the reflection coefficient of the SBH duct [37]. Mi et al. [38] has found that the open-end SBH can effectively reduce the reflection and transmission of sound waves. Su et al. [39] revealed that the mechanism of sound waves capture and acoustic streaming effects in the SBH terminate. It is pointed out that the distribution of particle velocity and sound pressure in the SBH terminate is uniform for low-frequency sound wave, resulting in weak sound absorption.

The aforementioned investigations indicate that the SBH duct, similar to the VBH, barely absorbs sound waves in low-frequency range. In short, there is also a cut-on frequency f_{cut-on} for SBH. Given that the continuous lumped admittance is achieved approximatively by means of the cavities separated by a finite number of rings [34,35], the SBH duct has one more factor than the VBH that inevitably affects the cut-on frequency: the depth of the cavity. Thus the determination of SBH's cut-on frequency is not only influenced by the terminal truncation, but also other structure parameters, including the ring number and thickness controlling the depth of the cavity. However, it is difficult to insert numerous thin rings in an actual SBH duct. Therefore, for the materialization of the SBH duct, the high cut-on frequency limits its application. To tackle this problem, a duct embedded periodic SBHs was presented by Mi et al. [40] to generate an attenuation band provided by the Bragg scattering effect in low-frequency range. This ingenious strategy greatly reduces the dependence of the SBH duct on the ring number. In addition, Chua et al. [41] incorporated the SBHs into truss lattices, and experimentally demonstrated that the proposed design achieves broadband sound absorption from 1000 to 6300 Hz. Despite these efforts, there are still few researches on improving the performance of the SBH-based duct for low-frequency sound absorption.

In a nutshell, it is recognized that effective sound absorption performance of the SBH duct relies on sound wave concentration due to slow wave effect and then abundant dissipation of concentrated sound energy. However, the damping and acoustic resistance of air is considerably weak, resulting in the failure of the SBH duct to obtain matched acoustic resistance in low-frequency range. Thus it will be a worth attempt to further enhance the sound energy dissipation effect of SBH in virtue of other sound absorption mechanisms. Based on this idea, the

micro-perforated panel (MPP) was selected as the energy dissipative component for this study. The MPP can provide sufficient acoustic resistance because the perforated holes with sub-millimeter diameter effectively intensify the friction caused by moving air, which renders the MPP as ideal sound absorption device [42]. Note that Zhang et al. [43,44] are the first to combine the MPP and SBH. In his study, a conical MPP was installed as the inner liner into a linear SBH duct, which exhibited excellent broadband sound absorption performance. Based on his work, in a recent research, Liang et al. [45] embedded a set of coiled resonators cavities to further broaden sound absorption. However, it should be pointed out that the inextirpable terminal truncation of the conical MPPs would incur the destruction of low-frequency sound absorption of this design [43]. Hence other forms of the MPP need be envisaged. It should be noticed that since the MPP have received much attention, various multi-layered or parallel-arranged MPPs have been proposed and proved the more effectiveness of these devices in low-and-medium-frequency sound absorption than the single-layered MPP [46–49]. Thus we grasp ideas from the multi-layered MPPs (MLP) system and envisage the combination with the SBH duct to sharpen its low-frequency performance. For brevity, we hereafter refer it as the MLPSBH duct. Meanwhile, due to the effective performance of the SBH duct in high-frequency range, the MLPSBH duct is also expected to provide broadband sound absorption, and promisingly overcome the hurdle of traditional resonant absorbers, whose absorption bandwidths are usually narrow.

Generally speaking, the purpose of this paper is to improve the defects of the SBH in low frequency sound absorption. Motivation for the employ of the multi-layered MPPs is to take advantage of its high acoustic resistance characteristics to rectify the mismatched acoustic resistance of the SBH duct in low-frequency range. We will present superb broadband and low-frequency sound absorption performance of the proposed device, which can be regarded as a superior sub-wavelength absorber. Specifically, the thickness of MLPSBH is only 1/15 of the wavelength corresponding to the cut-on frequency, and sound waves beyond the cut-on frequency can be effectively absorbed. This means that installation can be carried out in a narrow space, whereas traditional sound absorbers, such as Helmholtz resonators, rely on large cavities to achieve low-frequency sound absorption. The property would be conducive to the application of the MLPSBH duct.

The organization of the paper is as follows. In Section 2, we first illustrate the structure of the MLPSBH duct and develop the transfer matrix method (TMM) for sound absorption coefficient, which is verified by finite element method. In Section 3, several existing SBH absorbers are compared with the MLPSBH duct in order to detail the mechanisms of the broadband and low-frequency sound absorption. Various mentioned structural parameters and the influence of them to sound absorption are also discussed in this section. In Section 4, experiments are carried out to validate the accuracy of the TMM. The conclusions close the paper in Section 5.

2. Theoretical modeling of the MLPSBH duct

2.1. Model description of the MLPSBH duct

The proposed absorber combined with multi-layered MPPs and a SBH duct is illustrated in Fig. 1. The device is filled with air, and the solids inside it are assumed to be rigid. As suggested in [34], the basic SBH duct consists of a set of rings with varying inner radius, which are separated by cavities with same depth. The inner radius of rings follows a power law variation

$$r(x) = \frac{R - r_{end}}{L^n} |x|^n + r_{end} \quad (1)$$

where R is the radius of the waveguide; r_{end} is the residual radius; L is the length of the SBH duct; x is the longitudinal position of the ring; n is the

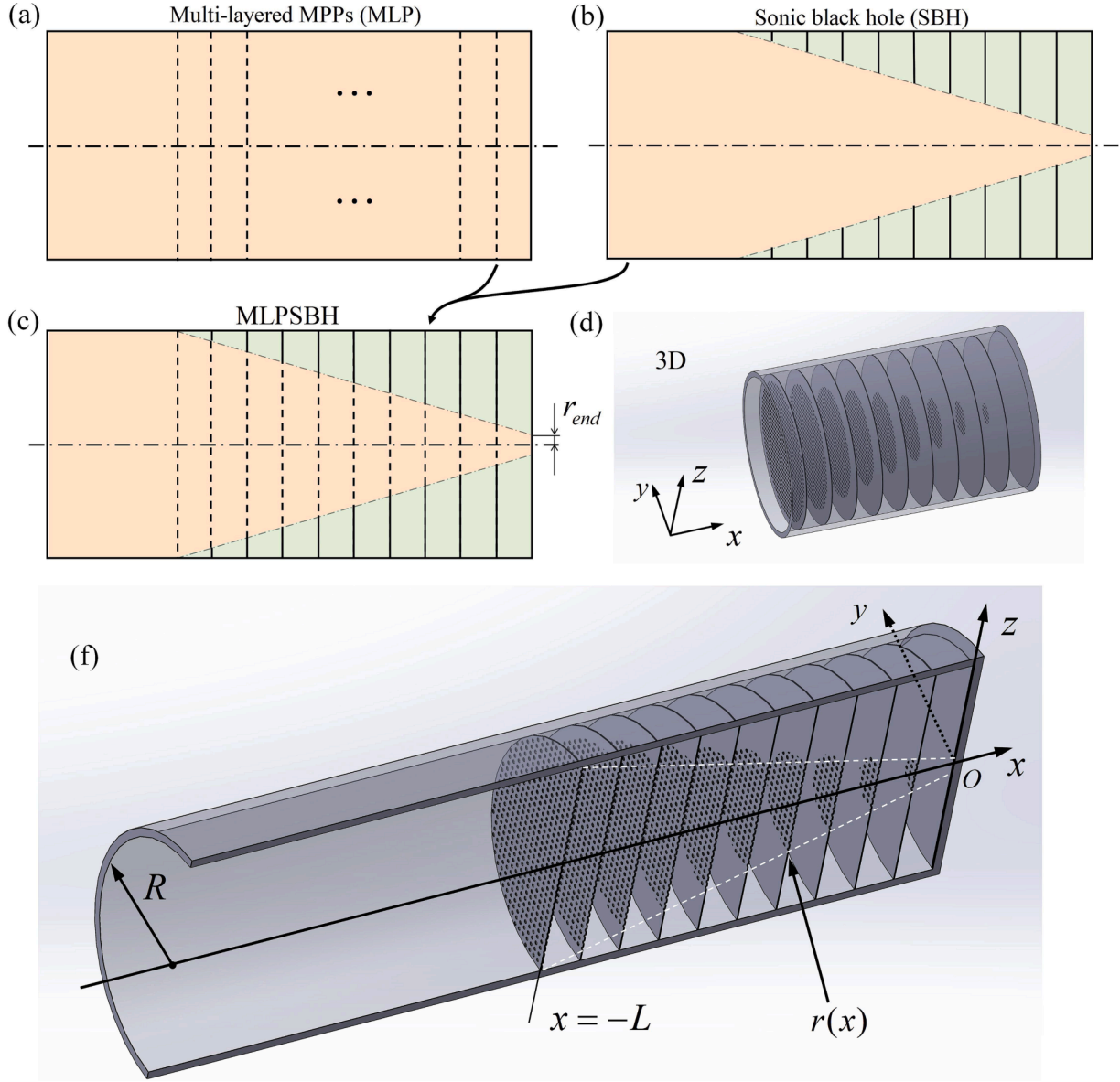


Fig. 1. (a) Multi-layered MPPs (MLP); (b) Sonic black hole (SBH); (c) Merged into the MLPSBH duct; (d) 3D view of the duct; (f) Profile of the duct. The inserted rings have a power-law decaying inner radius, which is equal to the outer radius of the corresponding MPP.

descending power of the inner radius of the rings. Noted that ideally the inner radius of the last ring will decrease to 0. However, the practical SBH duct could be truncated in the longitudinal direction due to the manufacturing limitation, which the radius of the truncation position is the residual radius. We focus on the cases with $n = 1$ (linear MLPSBH duct) and $n = 2$ (quadratic MLPSBH duct) in this paper.

The inside of each ring is mounted with a circular MPP. The size of each ring simultaneously governs the outer radius and thickness of the corresponding MPP. The acoustic impedance of the MPP consists of a resistance part and a reactance part, which can be achieved by Maa's theoretical formula [42]

$$Z_{MPP} = R_{MPP} + j\omega M_{MPP} \quad (2a)$$

$$R_{MPP} = \frac{32\rho\mu t}{\sigma d^2} \left[\sqrt{1 + \frac{K^2}{32}} + \frac{\sqrt{2}K}{8} \frac{d}{t} \right] \quad (2b)$$

$$M_{MPP} = \frac{\rho t}{\sigma} \left[1 + \frac{1}{\sqrt{9 + \frac{K^2}{2}}} + 0.85 \frac{d}{t} \right] \quad (2c)$$

where ρ is the density of air; μ is the kinematic viscosity of air; d is the diameter of the perforated holes; t is the thickness of the MPP; σ is the perforation ratio; $j = \sqrt{-1}$ is the imaginary unit; $K = \frac{d}{2} \sqrt{\omega/\mu}$; ω is the angular frequency.

2.2. Transfer matrix method of the MLPSBH duct

It has been explained by [35,38] that the transfer matrix method (TMM) is suitable for solving the reflection coefficient of the actual SBH duct. Based on the plane wave theory, the transitive relation of the state vectors (sound pressure p and volume velocity u) at the both ends of arbitrary one-dimensional acoustic model can be expressed by the transfer matrix. One of the important task in this section is to derive the TMM for the sound absorption coefficient of the MLPSBH duct. Next, we

will detail the elements of the transfer matrix of the MLPSBH duct with N rings according to its profile sketch, as shown in Fig. 2a.

A feature unit of the MLPSBH duct is shown in Fig. 2b, which can be divided into three acoustic regions, where Region 1 belongs to the MPP, and Region 2 belongs to the conical cavity, and Region 3 belongs to the toroidal cavity surrounding the conical cavity. The parameters $h_{r,i}$, $h_{r,i+1}$, d_c contained in the i -th feature unit and their meanings are illustrated in Fig. 2b. $h_{c,i}$ is the mean of the inner radii of the adjacent rings, which is used to represent the radius of the discretized conical cavity. Each acoustic region has a transfer matrix, in the case of the i -th feature unit, which can be represented by T_1^i , T_2^i , T_3^i respectively. Thus transitive relation of the state vectors $(p, u)^T$ at both ends of the i -th feature unit can be expressed as

$$\begin{aligned} \begin{bmatrix} p_i \\ u_i \end{bmatrix} &= T_1^i T_2^i T_3^i \begin{bmatrix} p_{i+1} \\ u_{i+1} \end{bmatrix} \\ &= \begin{bmatrix} 1 & \frac{Z_{MPP}}{S_{MPP}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos kd_c & j \frac{\rho_0 c_0}{S_i} \sin kd_c \\ j \frac{S_i}{\rho_0 c_0} \sin kd_c & \cos kd_c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_{cav}^i & 1 \end{bmatrix} \begin{bmatrix} p_{i+1} \\ u_{i+1} \end{bmatrix} \\ &= T_u^i \begin{bmatrix} p_{i+1} \\ u_{i+1} \end{bmatrix} \end{aligned} \quad (3)$$

where T_u^i is the transfer matrix of the i -th feature unit; Z_{MPP} is the acoustic impedance of the MPP, as referred to Eq. (2); $S_{MPP}^i = \pi h_{r,i}^2$ is the area of the MPP; $S_i = \pi h_{c,i}^2$ is the cross-sectional area of the Region 2; $k = \omega/c_0$ the acoustic wavenumber; Y_{cav}^i is the acoustic admittance of the Region 3, which can be achieved by plane wave method or lumped equivalent method. For the toroidal cavity in the SBH duct, the lumped admittance has higher accuracy [44], which can be expressed as

$$Y_{cav}^i = j \frac{k}{\rho_0 c_0} V_{cav}^i \quad (4)$$

where V_{cav}^i is the volume of the Region 3

$$V_{cav}^i = \pi d_c (R^2 - h_{c,i}^2) \quad (5)$$

The MLPSBH duct is composed of multiple feature units, so the transfer matrix of the entire MLPSBH duct can be characterized by means of successive products matrices

$$\begin{bmatrix} p_0 \\ u_0 \end{bmatrix} = T_u^1 T_u^2 \dots T_u^N \begin{bmatrix} p_N \\ u_N \end{bmatrix} = T \begin{bmatrix} p_N \\ u_N \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{bmatrix} p_N \\ u_N \end{bmatrix} \quad (6)$$

where T is the transfer matrix of the overall MLPSBH duct; $(p_0, u_0)^T$ is the state vector at the entrance and $(p_N, u_N)^T$ is at the terminate.

For the duct terminated with a rigid wall, the volume velocity at the terminate is zeros $u_N = 0$. And then the specific acoustic impedance can be obtained by Eq. (6). Note that the volume velocity should be converted to the particle velocity. The specific acoustic impedance of the MLPSBH duct can be expressed as

$$Z_{MLPSBH} = \frac{S}{\rho_0 c_0} \frac{p_0}{u_0} = \frac{S}{\rho_0 c_0} \frac{T_{11}}{T_{21}} \quad (7)$$

where $S = \pi R^2$ is the cross-sectional area of the duct.

The reflection coefficient at the entrance is obtained by Eq. (7)

$$R_{coe} = \frac{Z_{MLPSBH} - 1}{Z_{MLPSBH} + 1} \quad (8)$$

Finally, the sound absorption coefficient of the MLPSBH duct at normal incidence is given by

$$\alpha = 1 - |R_{coe}|^2 = \frac{4 \operatorname{Re}(Z_{MLPSBH})}{(\operatorname{Re}(Z_{MLPSBH}) + 1)^2 + \operatorname{Im}(Z_{MLPSBH})^2} \quad (9)$$

where $\operatorname{Re}(Z_{MLPSBH})$ is the specific acoustic resistance of the MLPSBH duct and $\operatorname{Im}(Z_{MLPSBH})$ is the specific acoustic reactance. When $\operatorname{Im}(Z_{MLPSBH}) = 0$, the absorber reaches resonance; when $\operatorname{Re}(Z_{MLPSBH}) = 1$, the absorber obtains matched acoustic resistance. This is the optimal impedance of a sound absorber, where the incident sound waves will be completely absorbed without reflection. Noted that the analysis frequency is limited by the plane wave assumption, as depicted in Eq. (10).

$$f \leq \frac{0.29 c_0}{R} \quad (10)$$

3. Numerical results of the MLPSBH duct

3.1. Sound absorption coefficient of the MLPSBH duct and validation

The precision of the TMM derived in the Section 2 is next validated by the FEM. Now let us introduce some significant details on the configuration of the MLPSBH duct. Numerical calculations of the sound absorption coefficient of the MLPSBH duct are conducted with the following parameters: $R = 30\text{mm}$, $L = 100\text{mm}$, $N = 10$, $r_{end} = 1\text{mm}$, $d = 0.2\text{mm}$, $t = 0.2\text{mm}$ and $\sigma = 0.04$. The counterpart contains a total of 10 combination of rings and MPPs. The depth of the cavity between two adjacent rings is 10 mm. Air damping is considered in the system by means of a complex sound speed $c = 340(1 + 0.05j)$, which roughly estimates the thermal-viscosity loss of the air [38]. Typically, the finite element model of the MLPSBH duct was established to verify the accuracy of the TMM, which was simulated by commercial software COMSOL. The MPPs are considered as the impedance surfaces in the FEM, whose acoustic impedance can be obtained by Eq. (2). The FEM models are meshed with tetrahedral elements, and the maximum mesh size was constraint to less than 1/6 of the minimum analysis wavelength. The transfer function method is employed to obtain the sound absorption coefficient in the FEM. In the numerical validation, we simultaneously calculated the sound absorption coefficient of the linear and quadratic MLPSBH ducts. The simulated results by the TMM and the FEM are shown in Fig. 3.

Now we focus on the precision of the TMM. The cut-on frequency with $\alpha = 0.5$ adopted in [43] is selected as the evaluation on the sound absorption performance. Specifically, the index means that more than 50 % of the sound energy can be absorbed above the cut-on frequency, as

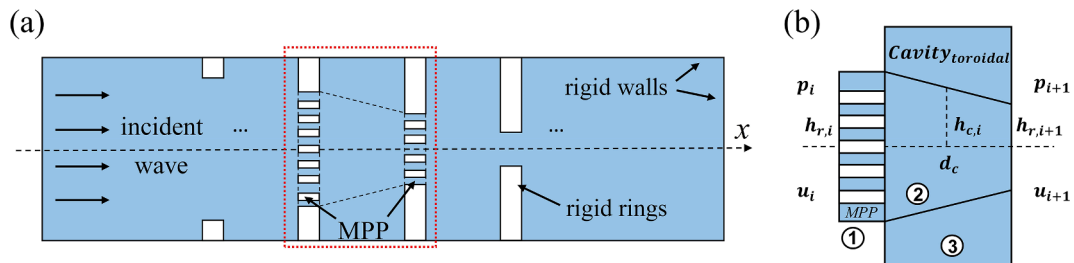


Fig. 2. (a) Profile sketch of the MLPSBH duct; (b) Feature unit.

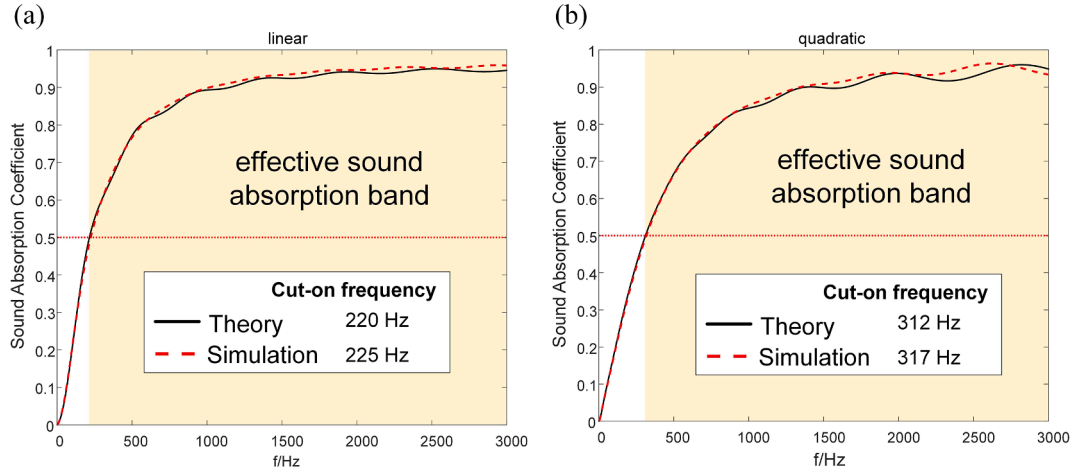


Fig. 3. Sound Absorption Coefficients of the MLPSBH ducts: (a) linear; (b) quadratic.

marked by the dotted line in Fig. 3. The cut-on frequencies of the linear MLPSBH duct achieved by the TMM and FEM are 220 Hz and 225 Hz, respectively, and those of the quadratic case are 312 Hz and 317 Hz, respectively. The relative errors with respect to the cut-on frequencies of the linear and quadratic MLPSBH duct, which can be got by $\varepsilon = |f_{FEM} - f_{TMM}|/f_{FEM} \times 100\%$, are 2.22 % and 1.58 %, respectively. Meanwhile, it is compared from Fig. 3 that the sound absorption coefficient obtained by two methods match well and there is only a tiny discrepancy in the high-frequency range. Thus it can be concluded that the precision of the TMM is validated. We note that for the linear MLPSBH duct, the cut-on frequency of 225 Hz corresponds to a wavelength $\lambda = 1.51\text{m}$. With $L = 100\text{mm}$, the length of the linear MLPSBH is about $\lambda/15$, which can be regarded as a sub-wavelength absorber. Such a low cut-on frequency of the absorber is partly attributed to the slow wave effect of the SBH duct. The effective attenuation of sound speed makes it take more time for sound waves to pass through the SBH duct, which can be considered as a disguised lengthening of the structure. The characteristic is not possessed by the traditional acoustic absorbers.

Comparing the cut-on frequencies of the linear and quadratic MLPSBH in Fig. 3, a notable observation can be found: the cut-on frequency of the linear case is 92 Hz lower than that of the quadratic case. This means that the linear MLPSBH duct has better low frequency performance. Previous researches [35,38] have indicated that the quadratic SBHs are more likely to trigger the acoustic black hole effect than the linear cases. However, the MLPSBH duct presented in this paper exhibits the opposite trend, but it is not difficult to understand that the area of the MPPs, for the quadratic case, is smaller than that in the linear case. Take the 10-th ring for example, the MPP radius of the quadratic MLPSBH is only 0.4 mm, while that of the linear case is 3.9 mm. To investigate the effect of the MPPs on the sound energy dissipation, the sound intensity and total power dissipation of two cases at 255 Hz,

which is a low frequency and near both the cut-on frequencies, are simulated by the FEM, as shown in Fig. 4. The sound intensity represents the concentration of the sound energy, and the total power dissipation represents the energy dissipation. It can be observed that the sound energy with $f = 255\text{Hz}$ in the linear case is mainly concentrated and dissipated near the 10-th ring; while in the quadratic case, it needs to be advanced to the 8-th ring, where the MPP can obtain the action area close to the linear case. This contrast indicates that the MPP has a positive effect on the sound absorption of the MLPSBH duct in the low-frequency range.

3.2. Comparison of several SBH-based absorbers

In this section, from a comparative perspective, several different SBH-based absorbers will be scrutinized to elaborate the sound absorption mechanism of the MLPSBH duct, with a focus on the role of the MPPs in low-frequency range. Three SBH-based absorbers are simulated below. The first is the MLPSBH duct presented in this paper; and the second is the SBH duct without modification; and the last is the SBH duct with the inner liner of a conical MPP, which is presented in [43] and referred as CMPSBH for brevity. The schematic models of the last two SBH absorbers are shown in Fig. 5. In comparison simulation, the significant parameters are ensured to be same with the CMPSBH investigated in [43], including the length of the absorbers $L = 300\text{mm}$, the number $N = 10$ and thickness $t = 0.2\text{mm}$ of rings, the outer radius of the ducts $R = 30\text{mm}$, and the residual radius $r_{end} = 1\text{mm}$. Meanwhile, the structural parameters of the MPPs are as described in the previous section, and the acoustic impedance of the MPPs are obtained by Eq. (2). It should be noticed that the FEM is adopted in the comparison simulation due to the inapplicability of the TMM to obtain the sound absorption coefficient of the CMPSBH duct.

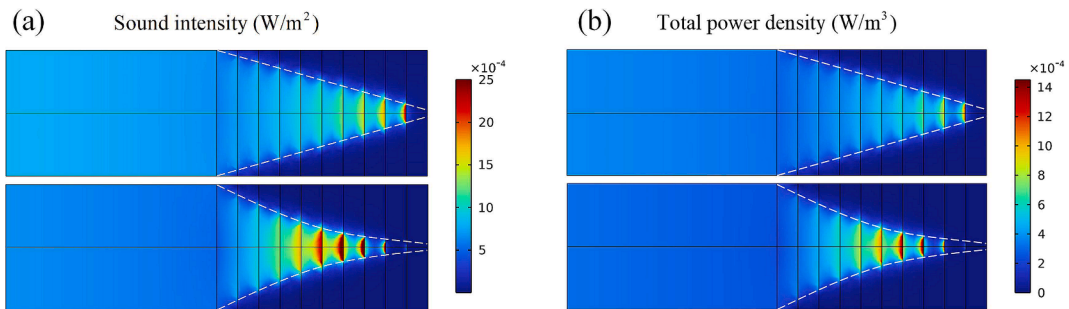


Fig. 4. Sound energy concentration and dissipation of the MLPSBH ducts: (a) Sound intensity; (b) Total power density. The analysis frequency is 255 Hz. The above figures are the results of the linear case and the below are the quadratic case.

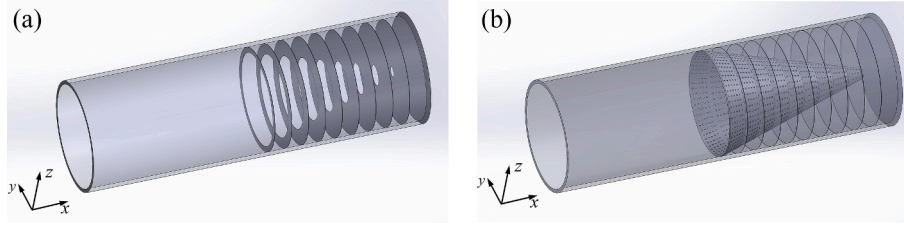


Fig. 5. Schematic models for comparison: (a) SBH duct; (b) CMPSBH duct.

The sound absorption coefficient of three SBH-based absorbers are shown in Fig. 6. As shown in Fig. 6, the high-frequency sound waves can be almost completely absorbed by the SBH duct. Such excellent high-frequency characteristic is perfectly retained by the MLPSBH and CMPSBH ducts, which is prerequisite for the realization of ultra-broadband sound absorption. Now we focus on the sound absorption performance of three absorbers in low-frequency range. In order to clearly demonstrate the respective advantages of the SBH and MPP, the perfect absorption cut-on frequency with $\alpha = 0.9$ has been added in the comparison, above which more than 90 % of the sound energy can be absorbed. The perfect cut-on frequency of the MLPSBH duct is 324 Hz, which is modestly lower than that of the CMPSBH duct. Both are far lower than the perfect absorption cut-on frequency, 1200 Hz, of the CMPSBH duct. It can be found that the absorption coefficient of the CMPSBH duct appears a jump and reaches resonance when the analyzed frequency decreases to 189 Hz. Although the CMPSBH duct shows better near this frequency, the MLPSBH duct outcompetes in the lower frequency range below 143 Hz. From the detail in Fig. 6, the lowest cut-on frequency of effective absorption with $\alpha = 0.5$ belongs to the MLPSBH duct, followed by the CMPSBH and the SBH without modification, representing 127 Hz and 270 Hz respectively. In other words, in the high-frequency range, the effective sound absorption performance of the SBH duct can be relies on sound wave concentration due to slow wave effect and then abundant dissipation in virtue of air damping. While the frequency is low where the dissipation caused by damping is rather weak, the MPPs begin to play a leading role instead of SBH in sound absorption. In summary, the interplay between the SBH and MPP effects generates the excellent broadband and low-frequency sound absorption of the MLPSBH duct. Next work will explore the respective roles of the

SBH and MPP play in the sound absorption.

3.2.1. Slow wave effect of the SBH duct

For an intuitive evaluation, a transient analysis was performed on three SBH-based ducts to simulate the propagation process of sound waves as passing through. The frequency of the incident wave is defined as 335 Hz, which coincides with the perfect absorption cut-on frequency of the MLPSBH and CMPSBH ducts. The propagation process of the sound waves in the ducts have been recorded by four signal probes, which can be seen in Fig. 7(a). Meanwhile, the first peaks of the sound pressure in four signal probes are linked to roughly depict the propagation time of the sound waves passing through the ducts. For comparison, the same simulation was carried out on a regular duct without modification and the result is shown in Fig. 7(b). Since the sound speed will not change in the regular duct, the red line marked in Fig. 7(b) is straight. The distance between probe 1 and probe 4 is $\Delta l = 0.27\text{m}$, while the propagation time of the sound wave is $\Delta t = 0.8\text{ms}$. So the sound speed in the regular duct is about $\Delta l / \Delta t = 337.5\text{m/s}$, which is consistent with the sound speed in air. However, the propagation time of the sound wave passing through the SBH duct is increased to $\Delta t = 1.783\text{ms}$. In addition, the slope of the dotted line in Fig. 7(c) gradually drops down, indicating that the sound waves come to slow down when they entering in the SBH duct. Previous researches have also pointed out that the sound speed decreases axially in the SBH duct [35,38], which clearly depicts the slow wave effect of it. Especially, for the MLPSBH and CMPSBH ducts, the propagation times of the sound waves are 1.834 ms and 1.817 ms respectively, roughly equal to that for the SBH duct. The phenomenon indicates that the slow wave effect of the SBH duct is inherited perfectly by the MLPSBH and CMPSBH ducts. The effect drives

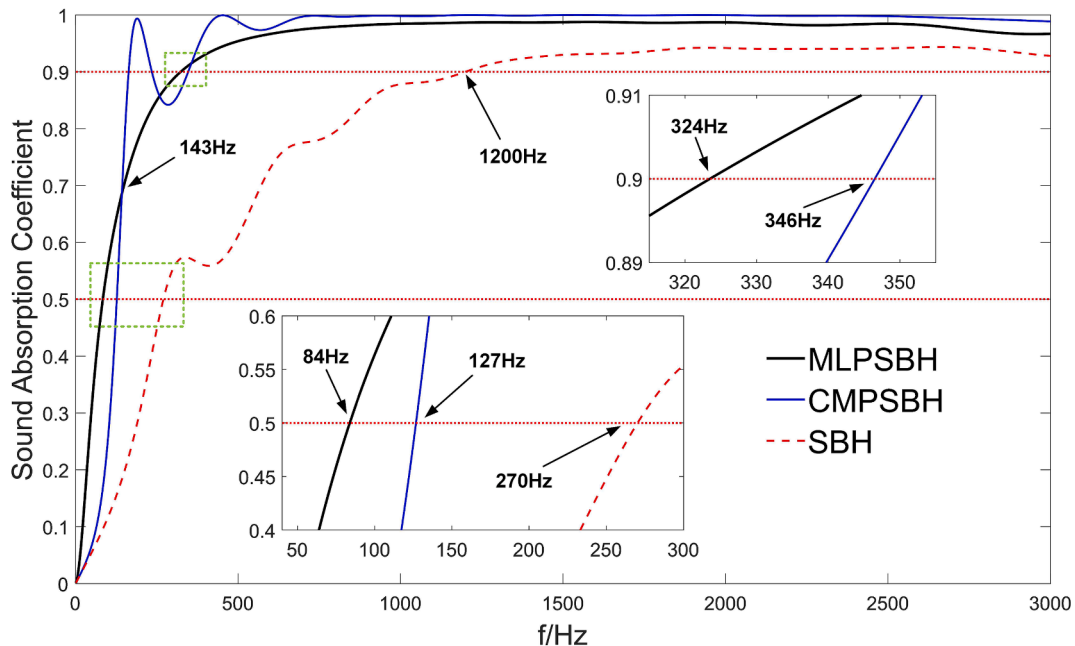


Fig. 6. Sound absorption coefficients of several SBH absorbers. With $N = 10$, $L = 300\text{mm}$, $R = 30\text{mm}$ and $r_{\text{end}} = 1\text{mm}$, the results are obtained by the FEM.

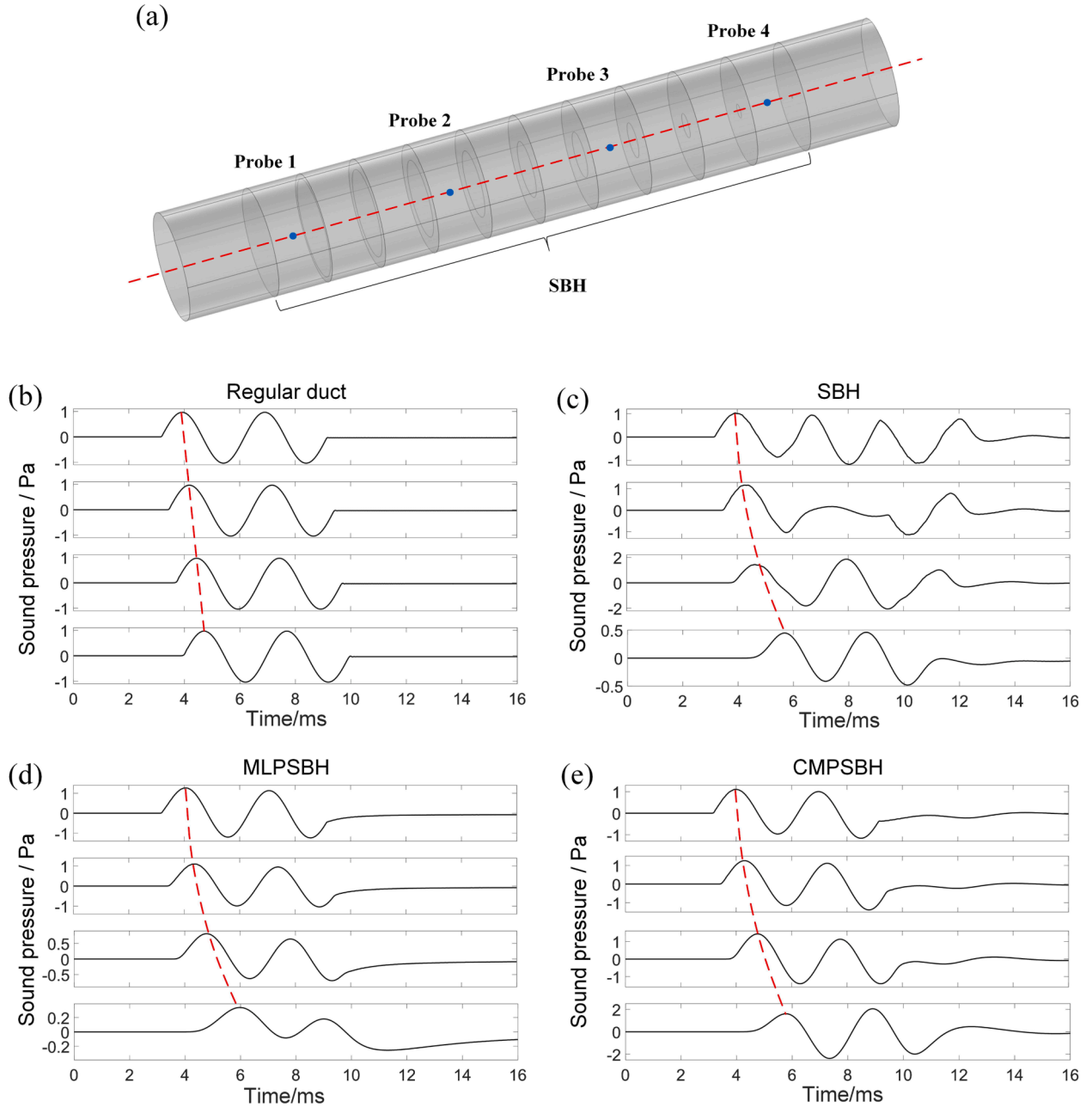


Fig. 7. Propagation process of the sound waves passing through the ducts: (a) positions of four signal probes; (b) the regular duct; (c) the SBH duct; (d) the MLPSBH duct; (e) the CMPSBH duct. The incident waves is excited with 335 Hz, which coincides with the perfect absorption cut-on frequency of the MLPSBH and CMPSBH ducts.

the sound wave to take longer to pass through the duct, thus can be considered as the elongation of the structural length. However, it can be found in Fig. 6 that the sound absorption coefficient of the SBH duct is only 0.574 at 335 Hz, which manifested that it is incompetent to only rely on the slow wave effect in low-and-medium-frequency range. Next work will exhibit the role that the MPPs play in sound absorption of the SBH-based ducts.

3.2.2. Compensation of the acoustic resistance by the MPPs

In order to further investigate the effect of the MPPs on the proposed absorber, the acoustic impedance of three SBH-based absorbers have been simulated, as shown in Fig. 8. According to Eq. (8), the reflection coefficient reaches the minimum when the specific acoustic resistance is 1. Thus the domain with specific acoustic resistance between 0.5 and 2 is assumed as the matched acoustic resistance domain, as the green shaded

region highlighted in Fig. 8. It can be observed that the acoustic resistance of all three absorbers is close to 1 in high-frequency range, among which the MLPSBH and CMPSBH ducts have the best matched acoustic resistance. Meanwhile, the acoustic reactance of three absorbers approaches zero rapidly after the transition in low-and-medium-frequency, which also explains the rationale why these absorbers have excellent sound absorption performance in high-frequency range. In the low-frequency we pay attention to, it can be found that the specific acoustic resistance of the SBH duct without modification deviates far from 1, leading to the deterioration of its property here. In fact, it was only after the discovery of this phenomenon that we proposed to improve the low-frequency mismatched acoustic resistance of the SBH duct by installing other high resistance devices. From Fig. 8(a), the acoustic resistance of the two absorbers equipped with MPPs is higher than that of the prototype SBH duct in low-frequency range, among which the MLPSBH duct

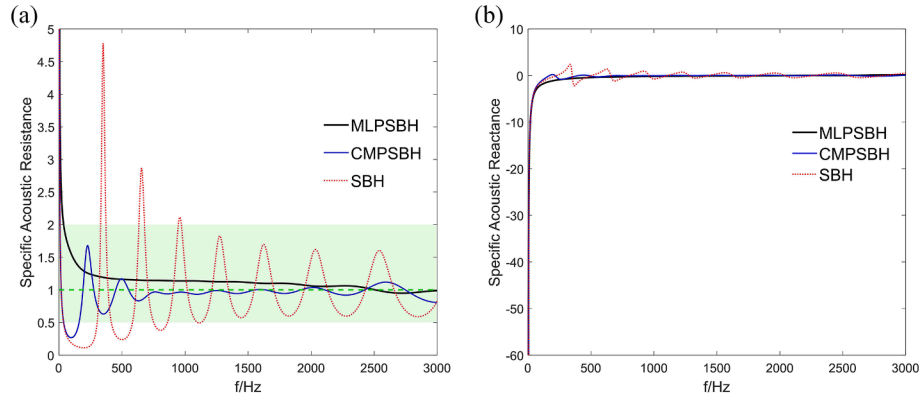


Fig. 8. Acoustic impedance of several SBH absorbers: (a) Specific acoustic resistance; (b) Specific acoustic reactance. The green region represents the matched acoustic resistance.

achieves the most obvious promotion. The acoustic resistance of the MLPSBH duct enters the matched region starting from 42 Hz and then converges monotonically towards 1, while the CMPSBH duct need delay to 158 Hz. For the SBH duct without modification, the acoustic resistance shows a significant oscillatory in low-and-medium-frequency range, and only when 916 Hz can it fully enter the matched acoustic resistance domain. This explains that the MLPSBH duct has the lowest cut-on frequency compared to the other two absorbers. The low-frequency matched acoustic resistance, coupled with the high frequency characteristics of the SBH, accomplishes the ultra-broadband and low-frequency sound absorption of the MLPSBH duct.

3.3. Analysis of structural parameters

Once the broadband and low-frequency sound absorption of the MLPSBH duct has been demonstrated in the previous section, this section will next focus on the influence of some significant parameters on the properties of the MLPSBH duct, especially the cut-on frequency. One of the incentives of the paper is to further expand the research of the SBH absorbers, so the parameters we pay attention to in this analysis belong to the common characteristics of the SBH absorbers, rather than the MPP. The structural parameters of the basic MLPSBH duct have been described at the beginning of the previous section.

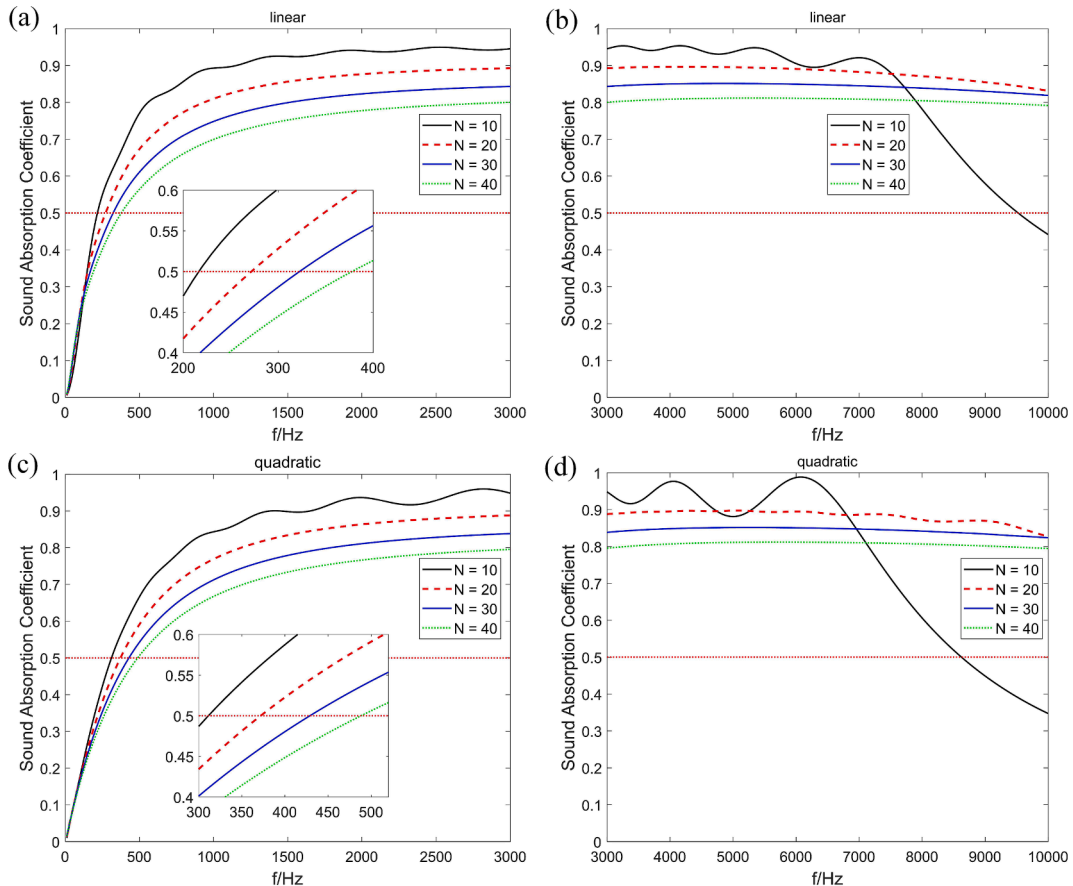


Fig. 9. Sound absorption coefficients of the MLPSBH ducts with different number of rings: (a) & (b) the linear cases; (c) & (d) the quadratic cases.

3.3.1. Number of rings

In the derivation of the governing equation of the SBH duct [34], the lumped admittance is a continuous function respect to x . The previous researches have revealed that if the materialization of the SBH absorber would retain the properties shown by the governing equation (i.e. the continuity of the lumped admittance), the rings inserted in the duct need to be abundant and thin enough [35]. It is also found that the reflection coefficient of the SBH absorber increases remarkably when the number of rings N reduces below 40. On account of the difficulty of installing many thin rings in a narrow space, the dependence on the number of rings limits the application of the SBH absorber. Therefore, it is worth exploration whether the MLPSBH duct can maintain the desirable sound absorption performance with the alteration of the ring number.

Fig. 9 gives the results of the linear and quadratic MLPSBH ducts with different ring numbers. As N decreases from 40 to 10, the cut-on frequencies of the linear and quadratic cases decline 161 Hz and 177 Hz, respectively. The marvelous information demonstrates that the MLPSBH duct guarantees excellent low-frequency sound absorption without abundant rings. Among previous researches, only periodic SBHs duct exhibits similar properties [40]. However, for the MLPSBH duct, the number of rings cannot reduce indefinitely. In order to detect this problem, we extend the analysis frequency to 10000 Hz. Although such high frequency is beyond the limitation of plane wave analysis, the trend of sound absorption performance still provides some valuable directions. From Fig. 9b, as N decreases to 10, the sound absorption coefficient of the linear case drops significantly after 7000 Hz. When the number of rings decreases uninterruptedly, the proposed absorber will destruct the high-frequency sound absorption effect of the SBH duct and gradually exhibit an oscillatory behavior, which is similar to that of series-arranged MPPs [50]. In order to further illustrate this phenomenon, we simulated the sound absorption of the linear MLPSBH duct with 5 rings, as shown in Fig. 10. Due to the distortion of the admittance Y_{cav} caused by the lack of adequate rings, the FEM is adopted to simulate this case. The result also shows a similar oscillatory behavior, and the dropping frequency is advanced to 4000 Hz. The conception of using multiple resonances to broaden the sound absorption band is considerably similar to the series-arranged MPPs [50]. In addition, for the lack of rings, the cavities behave as Helmholtz resonators with varying parameters. This allows the sound waves with different frequencies will be captured by the cavities with different depth, formally termed as acoustic rainbow trapping [51], which can be learned more in [52]. When the analysis frequency exceeds all the eigenfrequencies of the cavities, the sound absorption of the proposed device will drop dramatically. In this case, the absorber no longer retains the performance of the SBH duct. Despite the unsatisfactory sound absorption at high frequencies, the absorber still shows a superior broadband property in medium-low-frequency range. The discovery may provide new directions for the study of the absorbers with series and parallel arranged

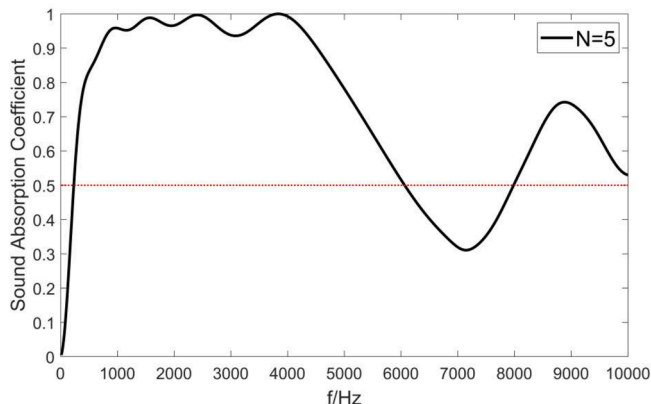


Fig. 10. Sound absorption coefficient of the MLPSBH duct with 5 rings.

MPPs.

3.3.2. Thickness of the ring

The effect of ring thickness on the sound absorption of the SBH absorber is not conclusive. For the prototype SBH duct, the sound absorption performance is not sensitive to the ring thickness [35]. However, a recent study found that the Bragg bandgaps of the period SBH duct can be widened by increasing ring thickness [40]. In addition, for the MLPSBH duct presented in the paper, the thickness of the ring is actually equal to that of the MPP, which also affects the sound absorption performance of the MPPs. Therefore, it is imperative to anatomize the influence of the ring thickness on the sound absorption of the MLPSBH duct.

To proceed, the number of rings has been fixed to $N = 10$ and the characteristics of the MPPs except thickness have been not changed. Fig. 11 shows the result of the linear and quadratic MLPSBH ducts with varying ring thickness. As the ring thickness decreases, for the linear case, the cut-on frequency shifts towards low frequency. The sound absorption coefficient curve of the absorbers with thin rings is almost overall higher than that of the absorbers with thick rings. The same conclusion can be given by the results of the quadratic case. It is not difficult to explain. The previous researches on the MPP absorbers have shown that the reduction of the MPP thickness will cause the shift of the effective band to the low frequency and the increase of the sound absorption coefficient [53]. Meanwhile, it should be noted that the number of rings and the length of the duct have been fixed, so that increasing the ring thickness will undoubtedly reduce the cavity length and weaken the continuity of the admittance. The underlying changes give rise to the impairment of the sound absorption with the increase of the ring thickness. Thus, the desired low-frequency sound absorption of the MLPSBH duct partly depends on the thin rings.

3.3.3. Length of the MLPSBH duct

Let us consider the effect of the structural length on the sound absorption of the MLPSBH duct. It is always hoped that the absorber can maintain the sound absorption performance with a narrow installation space, which is the motivation for extensive efforts to develop the sub-wavelength absorbers. Subsequently, the number and thickness of rings have been fixed to $N = 10$ and $t = 0.2\text{mm}$, respectively, and note that we have not varied the characteristics of the MPPs to separate their effects.

The sound absorption coefficients of the MLPSBH ducts with different structural length are plotted in Fig. 12. It can be seen that the length of the MLPSBH duct has a great influence on the sound absorption. As the absorber becomes longer, the cut-on frequency decreases rapidly. The cut-on frequency of the linear MLPSBH duct with $L = 200\text{mm}$ is only 127 Hz, which is a rather low frequency and the corresponding wavelength is roughly 13.4L. Similar transformation also happens to the quadratic case. It is preliminarily speculated that this phenomenon may be related to the cavity depth between two adjacent rings. Due to the invariant of the number of rings, the depth of each cavity obeys a multiple relation with the length of the duct. This reminds us that the wavelength of the passing sound wave is closely related to the cavity depth in traditional mufflers and absorbers. Many reactive mufflers utilize the characteristic to block the sound wave with a certain frequency. For instance, expansion segments are usually used in pipes to eliminate the sound waves with the wavelengths up to four times their length.

In order to verify the conjecture, the cut-on frequencies of the MLPSBH duct with different cavity depths are computed. Obviously, the cavity depth is determined by the length of the absorber L , the number N and thickness t of rings. To avoid the influence of the ring thickness on sound absorption of the MPPs, we have fixed the ring thickness and other structural parameters. Only L and N are selected in the investigation to analyze the impact on the cut-on frequency of the linear and quadratic MLPSBH ducts, as shown in Fig. 13. The red point in the figure

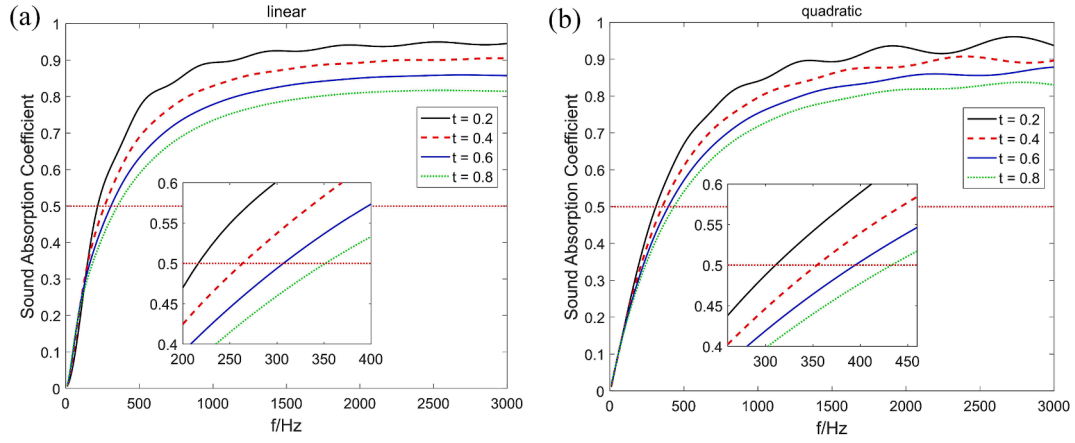


Fig. 11. Sound absorption coefficients of the MLPSBH ducts with different ring thickness: (a) the linear cases; (b) the quadratic cases.

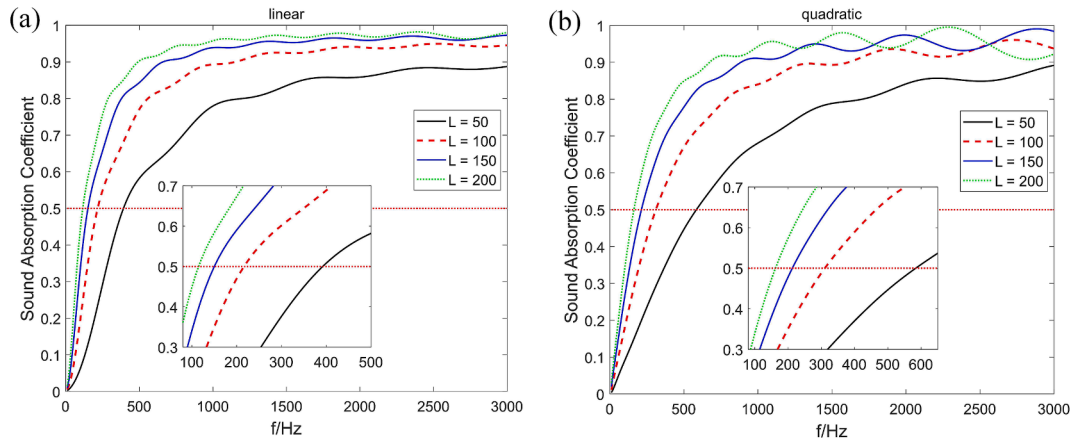


Fig. 12. Sound absorption coefficients of the MLPSBH ducts with different structural length: (a) the linear cases; (b) the quadratic cases.

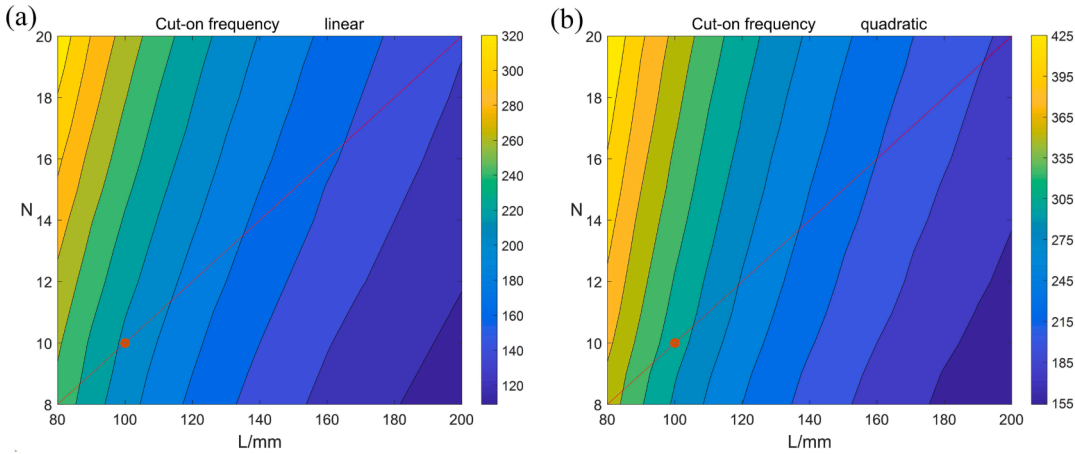


Fig. 13. Cut-on frequency versus the length of the duct L and the number of rings N : (a) the linear cases; (b) the quadratic cases.

represents the prototype absorber with $N = 10$ and $L = 100$ mm. The red line passes through the initial point, representing the cases with the cavity depth equal to the prototype absorber. The points on the lower right of the red line represent the cases with the cavity depth higher than the prototype, and the points on the upper left represent the opposite cases. For both the linear and quadratic cases, it can be observed that the data in the figures monotonically slide towards the minimum value in the lower right corner, which indicates that the MLPSBH duct with

larger cavity depth has lower cut-on frequency. The results are consistent with the previous perspective. But closer observations show that the trend is not proportional due to the difference of the cut-on frequencies on the red line. Noted that the data points on the red line share the same cavity depth. In addition, the isolines colored in black are steep rather than gentle, which depicts that the length of the MLPSBH duct has a greater influence on the cut-on frequency than the number of rings. Conclusively, the desired cut-on frequency of the absorber can be

obtained by modifying such parameters conveniently.

3.3.4. Influences of damping

As elaborated in the review, the purpose of the employment of the MPPs is to enhance the sound energy dissipation to assist low-frequency sound absorption. Naturally, one could wonder whether additional damping, such as porous fiber and glass cotton, is needed to raise the energy dissipation. It is pointed out in previous section that the damping is considered by means of a complex sound speed $c = 340(1 + \eta j)$, where η is the damping coefficient. In this section, the influence of the damping on the sound absorption has been explored for the linear and quadratic MLPSBH ducts, as shown in Fig. 14. The structural parameters have been fixed as $N = 10$, $L = 100\text{mm}$ and $t = 0.2\text{mm}$.

Fig. 14 shows that the sound absorption coefficient is almost independent on the damping. Compared with the prototype with $\eta = 0.05$, the cut-on frequency of the linear case with no damping increases only 4 Hz, and the sound absorption coefficient shows marginal fluctuation. Combined with the conclusion in Section 3.1, the insensitivity of the sound absorption to damping indicates that the MPPs play a leading role in the sound energy dissipation of the MLPSBH duct, which is consistent with the intention of the design. In fact, for the SBH structural proposed in the previous researches [35,40], heavy damping is required to give full play to its sound absorption performance, resulting in the demand of the combination with absorbent material for application to some extent. In contrast, sufficient sound absorption of the MLPSBH duct can be obtained without any other effort.

4. Experiment validation

To validate the accuracy of the simulated result, the sound absorption coefficients of the designed MLPSBH duct were measured by two-microphone transfer function method in an impedance tube. In order to ensure that the test sample can restore accurately the desired characteristics, the MLPSBH duct was manufactured in several segments, including the container, MPPs, separators and the limiting stopper. During installation, the separators and the MPPs were mounted into the container in sequence, and finally fixed by the limiting stopper, which was bolted with the container, as shown in Fig. 15. Noted that the inner radius of the last ring can be kept as millimeter-size in that the number of rings is not large. Thus the residual radius r_{end} is regarded as 0 in manufacturing. The rest parameters of the manufactured MLPSBH duct are consistent with the analyzed object in Section 3.1: $L = 100\text{mm}$, $R = 30\text{mm}$, $N = 10$, $d = 0.2\text{mm}$, $t = 0.2\text{mm}$ and $\sigma = 0.04$, where the detail of the MPPs can be seen in Fig. 15b. For the experiment, the final result was obtained as an average of five measurements. The testing process refers to the ISO 10534-2, as shown in Fig. 16a. Note that the minimum effective frequency is 50 Hz, which is subject to the physical limitation

of the loudspeaker.

The measured sound absorption coefficient is shown in Fig. 16b, which is compared with the result obtained by the TMM. The measured result is in good agreement with that of the simulation in general trend, and the cut-on frequency calculated by the TMM is only 6 Hz lower than the measured result, which verify the numerical simulation. The discrepancies between the two sets of results can be attributed to the machining errors and the neglect of the influence of the structural flexibility on the results in numerical simulation. Even so, it can be concluded from the above results experimentally and theoretically that the proposed absorber has excellent broadband and low-frequency sound absorption.

5. Conclusion

In this paper, we have proposed the design of the MLPSBH duct which possesses a broadband and low-frequency sound absorption. The proposed absorber is designed to have multi-layered MPPs inserted into the SBH rings with a power-law decaying inner radius. The transfer matrix method (TMM) has been deduced to analyze the sound absorption performance of the MLPSBH duct, which is validated by the finite element method (FEM) and well-designed experiment. For linear MLPSBH duct, the cut-on frequency of 0.5 absorption coefficient is 225 Hz, whose wavelength is 15 times the length of the duct. It also turns out that due to the larger action area of the MPPs, the linear MLPSBH duct shows the superior low-frequency performance than the quadratic case, reflected by the cut-on frequency of 100 Hz lower.

In order to interpret the SBH and MPP effects, the sound absorption and the acoustic impedance of several different SBH-based absorbers have been simulated and investigated. The results show that the MLPSBH duct inherits the slow wave effect of the SBH duct perfectly. In addition, the high acoustic resistance of the MPPs effectively compensates for the mismatched acoustic resistance of the SBH duct in low-frequency range. Combined the characteristic of the SBH and MPP effects, the MLPSBH duct achieves ultra-broadband and low-frequency sound absorption.

Based on the TMM, the influence of structural parameters on the cut-on frequency and absorption performance of the MLPSBH duct has been elaborated. When the rings are sufficient, the proposed absorber maintains broadband performance, otherwise its sound absorption becomes the characteristics of similar to the series-arranged MPPs. Compared with the number and thickness of rings, the length of the absorber is the main factor determining the cut-on frequency. A potential relationship has been also found that the absorber with larger cavity depth gains a lower cut-on frequency. The damping analysis shows that the MLPSBH duct can obtain the desired broadband sound absorption without the additional absorbent material.

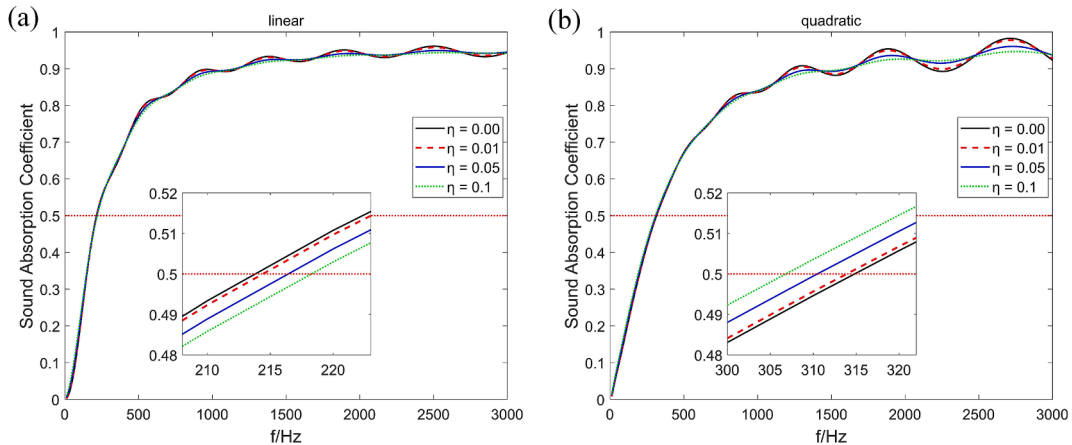


Fig. 14. Sound absorption coefficients of the MLPSBH ducts with different damping: (a) the linear cases; (b) the quadratic cases.

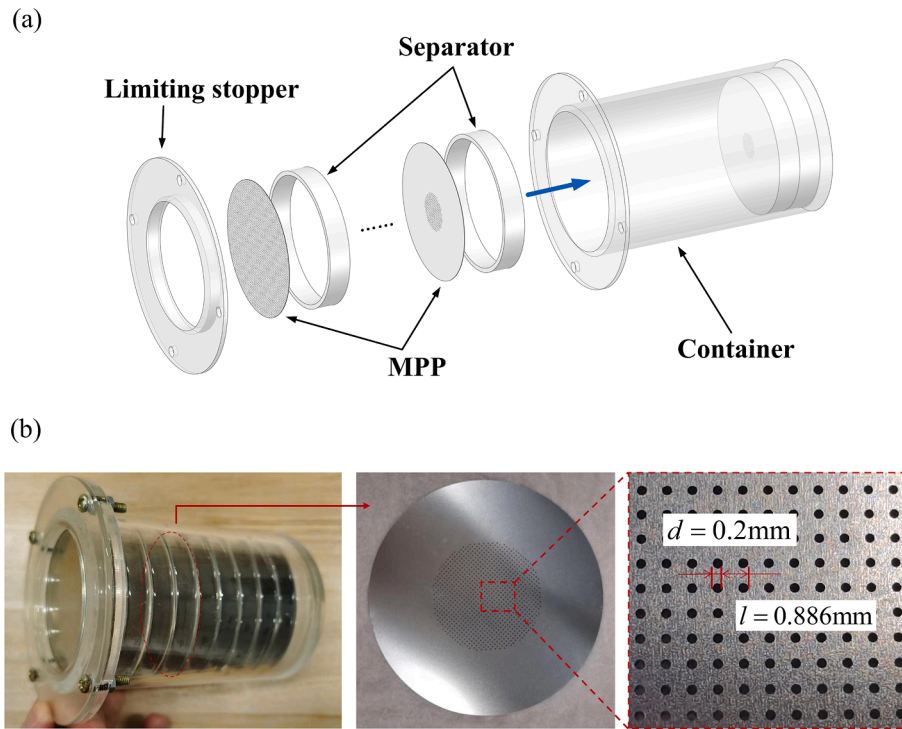


Fig. 15. (a) Assembly diagram of the MLPSBH duct; (b) Photo of the manufactured MLPSBH duct.

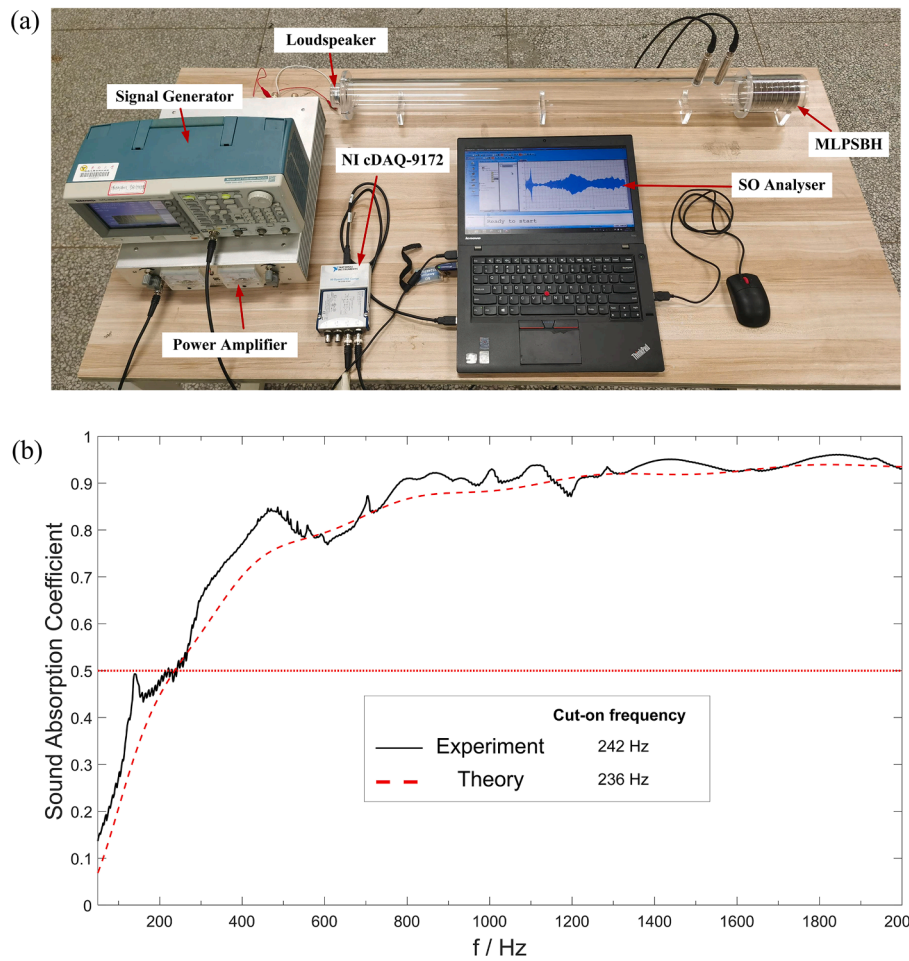


Fig. 16. (a) Test system of the sound absorption coefficient in an impedance tube; (b) Comparisons between the measured and theoretical results.

This paper proves that promoting the dissipation of sound wave is an effective method to improve the low-frequency property of the SBH duct. Thus incorporating other damping absorbent materials into the SBH-based absorbers would deserve exploration in the further. To conclude, the proposed design provides a promising concept for the applications of the SBH-based absorbers in noise control.

Funding

The National Natural Science Foundation of China (Grant No. 52275092 and Grant No. 52202478).

CRediT authorship contribution statement

Yunwei Chen: Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kangfan Yu:** Methodology, Software, Validation. **Qidi Fu:** Methodology, Software, Validation. **Jianrun Zhang:** Funding acquisition, Project administration, Supervision, Writing – review & editing. **Xi Lu:** Funding acquisition, Supervision. **Xiaofei Du:** Investigation, Supervision. **Xiaojuan Sun:** Funding acquisition, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

The authors gratefully acknowledge the support for this work by The National Natural Science Foundation of China (Grant No. 52275092 and Grant No. 52202478).

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